

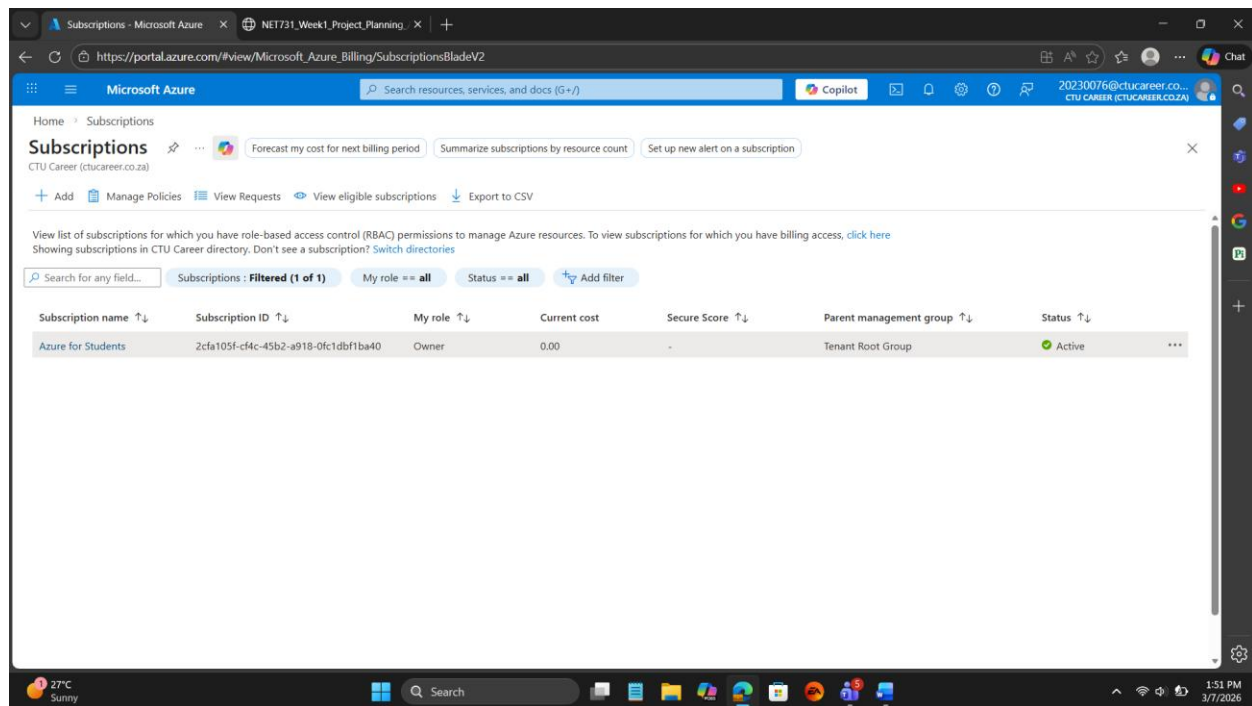
ONNEILE TLOUANA. 20230076. NET731. WEEK 1

This practical focused on planning and setting up a Microsoft Azure cloud environment for CTU Training Solutions. The objective was to identify business requirements, select appropriate Azure services, and organize resources using Azure Resource Groups.

Below are screenshots of

- Defined business requirements for CTU Training Solutions
- Identified Azure services required
- Azure Resource Group in Azure for Students
- Documented region, tags, and configuration

TASK 1



TASK 2

1.What type of organization is CTU Training Solutions?(size, industry, purpose)

-CTU Training Solutions is an institution of higher education and skills development.

-CTU is a private tertiary education provider that is registered with the department of higher education and training in South Africa.

-It is a medium sized organization that has an estimate of 200 to 500 employees.

-Its purpose is to upskill people, improve employability, offers diplomas, national qualification, short courses and international certifications in various fields such as information technology, engineering, etc.

2.What are the main IT needs of the organization?(consider: staff, students, data, applications)

- With staff needs, they will requires learning management systems, email platforms, collaboration tools, etc.

-Student needs would be, online learning systems, access to course materials, student portals, collaboration tools, authentication system.

-Data applications, learning management systems, collaboration tools and student information systems

3. Why would a cloud solution (Azure) be more beneficial than on-premises infrastructure for CTU?

-Scalability, cloud resources are able to scale accommodate high volumes of traffic

-Cost efficiency, virtualized resources are cheaper to run, maintain and azure has different subscription options

-Remote/hybrid learning supported as cloud platforms and be accessed anywhere in the country

4. What security and compliance requirements should be considered?

-Data protection for sensitive information by using access control and encryption

-IAM,MFA and RBAC

-Network security systems like firewalls, DDOS protection

5. What is the expected number of users who will access the cloud environment?

+ 5000 to 8000 full time and part time students across the country(12 campuses)

Academic and admin staff +- 1000

BIBLIOGRAPHY

-CTU Training Solutions (2024) About CTU Training Solutions. <https://www.itweb.co.za/office/ctu-training-solutions-pty-ltd/about>

-CTU Training Solutions (2024) Training provider profile. https://www.skillsportal.co.za/train/training_providers/view/ctu-training-solutions

-CTU Training Solutions (2024) CTU Campuses. <https://www.ctutrainig.ac.za/ctu-campuses/>

-CTU Training Solutions (2024) Where to study: CTU Training Solutions.

<https://www.careersportal.co.za/where-to-study/ctu-training-solutions>

-CTU Training Solutions (2024) Company profile. <https://za.linkedin.com/company/ctu-training-solutions-pty-ltd>

-Microsoft (2024) What is cloud computing? <https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-cloud-computing/> .

-Protection of Personal Information Act (2013) Protection of Personal Information Act 4 of 2013. <https://www.gov.za/documents/protection-personal-information-act>

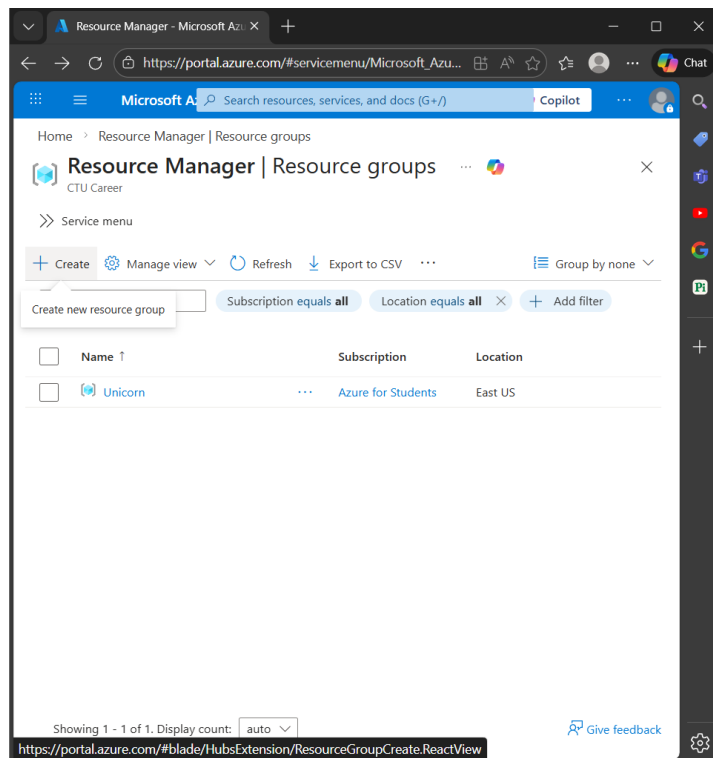
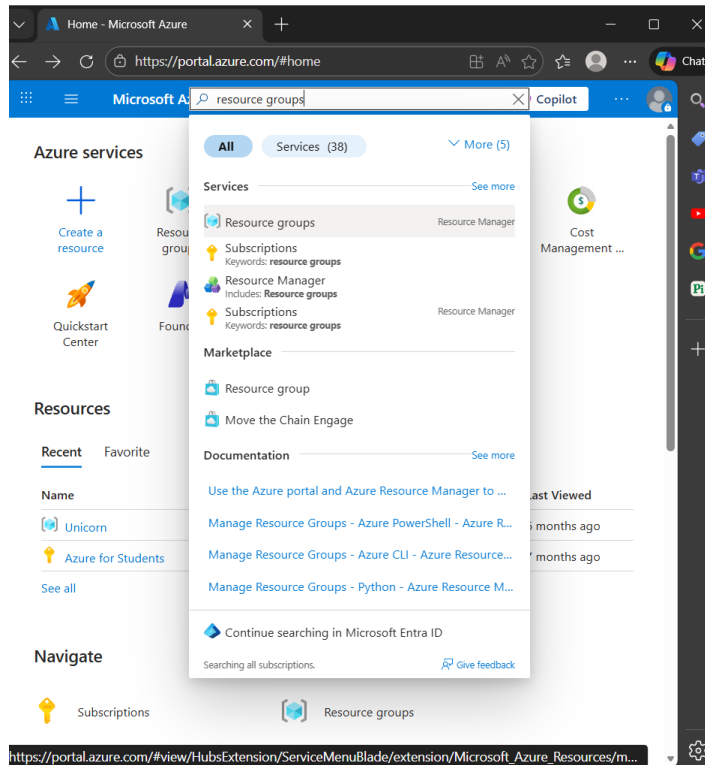
-National Institute of Standards and Technology (2020) Cloud Computing Overview. <https://www.nist.gov/itl/cloud>

TASK 3

Azure Service	Purpose / Why you need it	Week it is used
Resource Group	It is a container that is used to organize and manage azure resources	01
Virtual Network (VNet)	A private network in Azure	02
Virtual Machine (VM – B1s)	A cloud based computer used to run apps and services	03
Microsoft Entra ID	Manages user identities and secure access to Azure resources	04
Network Security Group (NSG)	Controls network traffic by allowing or blocking access to Azure resources.	05

Azure Storage Account	Stores files, backups, and application data securely in the cloud.	05
Microsoft Defender for Cloud	Provides security monitoring and threat protection for Azure resources	06
Azure Monitor	Tracks performance, logs and health of Azure resources	07
Azure Cost Management	Helps monitor and control cloud spending.	08

TASK 4



Create a resource group - Microsoft

https://portal.azure.com/#view/HubsExtension/ResourceGr...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription *

Azure for Students

Resource group name *

CTU-20230076-RG

Region *

(Africa) South Africa North

Previous

Next

Review + create

Create a resource group - Microsoft

https://portal.azure.com/#view/HubsExtension/ResourceGr...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name	Value	Resource
Project	NET731-Capstone	Resource group
Student	Onnelle Tlouana	Resource group
Module	NET731	Resource group
		Resource group

Previous

Next

Review + create

Resource Manager - Microsoft Az...

https://portal.azure.com/#servicemenu/Microsoft_Azure_Re...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Chat

Home > Resource Manager

Resource Manager CTU Career

>> Service menu

+ Create ...

Group by none

Filter for any field...

Subscription equals all Location equals all

+ Add filter

☐ Name ↑

Su

☐ CTU-20230076-RG

...

Az

☐ Unicorn

...

Az

Showing 1 - 2 of 2. Display count: auto

Give feedback

Notifications

More events in the activity log →Dismiss all

✓ Resource group created

×

Creating resource group 'CTU-20230076-RG' in subscription 'Azure for Students' succeeded.

Go to resour... Pin to da...

2 minutes ago

Resource Manager - Microsoft Az...

https://portal.azure.com/#servicemenu/Microsoft_Azu...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Chat

Home

Resource Manager | Resource groups CTU Career

>> Service menu

+ Create Manage view Refresh Export to CSV

Group by none

Filter for any field...

Subscription equals all Location equals all

+ Add filter

☐ Name ↑

Subscription

Location

☐ CTU-20230076-RG

...

Azure for Students

South Africa North

☐ Unicorn

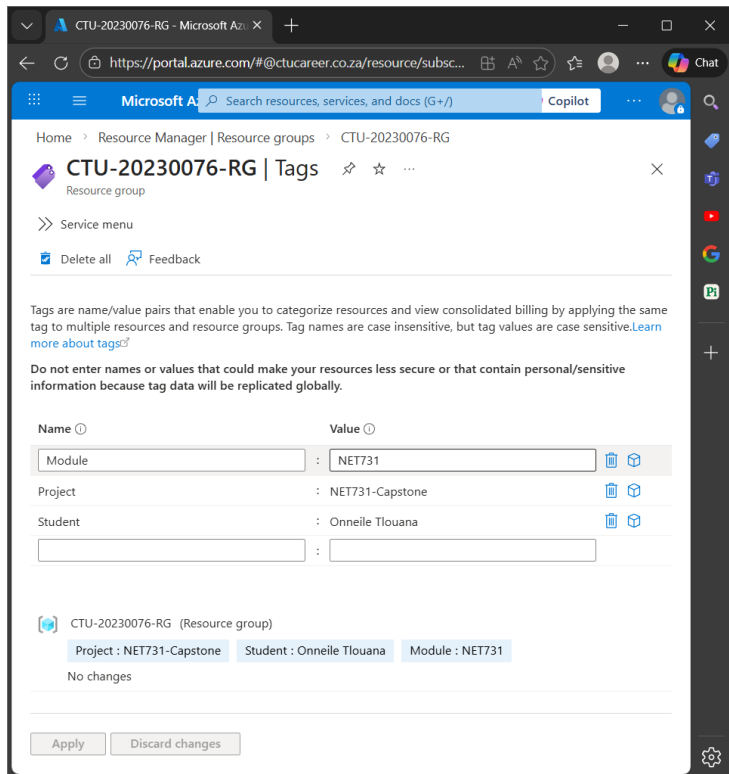
...

Azure for Students

East US

Showing 1 - 2 of 2. Display count: auto

Give feedback



REFLECTION

This practical improved my understanding of cloud planning, resource organization and Azure environment preparation.

SKILLS DEMONSTRATED

- Cloud Planning
- Azure Fundamentals
- Resource Group management
- Business Analysis